MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE COAXIAL, 50 OHMS, M17/128-RG400

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-17.

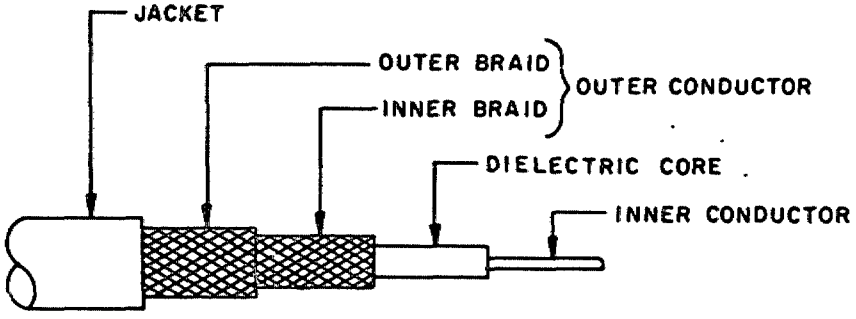


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Nineteen strands of silver-coated, copper wire at .008 inch each. Overall diameter: .0384 inch ±.0010.
Dielectric core	Type F-1: Solid extruded PTFE. Diameter: .116 inch ±.005.
Outer conductor	Double braid of AWG size 36 silver-coated wire. Diameter: .171 inch maximum.
Inner braid	Coverage: 94.8% nominal Carriers: 16 Ends: 7 Picks/inch: 11.5 ±10%
Outer braid	Coverage: 93.6% nominal Carriers: 16 Ends: 7 Picks/inch: 14.5 ±10%
Jacket	Type IX. Diameter: .195 inch ±.005.

FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage: 1,400 Vrms, maximum.

Operating frequency: 12.4 GHz, maximum.

Velocity of propagation: 69.5 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -55° to +200°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.91 ohm per 100 feet.

Elongation: 10 percent, minimum.

Engineering notes: This cable useful in general purpose, high temperature applications (see connector series "TNC", "BNC", and "SMA" per MIL-C-39012).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Adhesion of conductors:

Inner conductor to core: 4 pounds, minimum; 30 pounds, maximum.

Aging stability: Not applicable.

Stress crack resistance: +230° ±5°C; mandrel size seven times the jacket diameter.

Outer conductor integrity: Not applicable.

Dimensional stability: +200° ±5°C.

Inner conductor from core: .187 inch, maximum.

Inner conductor from jacket: .312 inch, maximum.

Bendability: Not applicable.

Weight: 5 pounds per 100 feet, maximum.

Electrical:

Test frequency: 50 MHz to 12.4 GHz.

Spark test: 2,000 Vrms, +25 percent, -0 percent.

Voltage withstanding: 3,000 Vrms, minimum.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 Vrms, minimum.

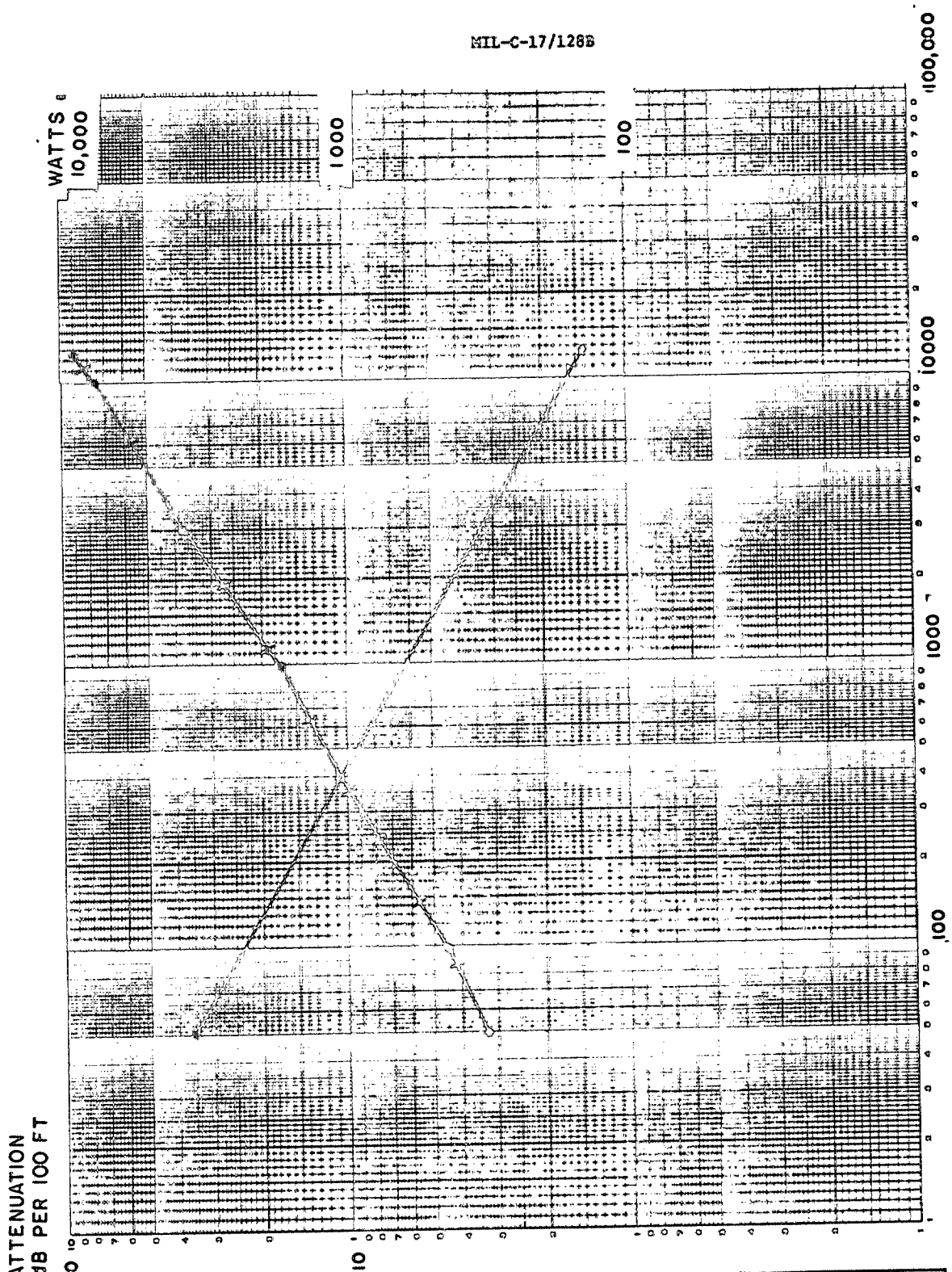
Characteristic impedance: 50 ±2 ohms.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32 pF per foot, maximum.

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Freq MHz	Attenuation dB
50	3.2
100	4.5
400	10.5
1,000	17
3,000	38
10,000	78
12,400	90

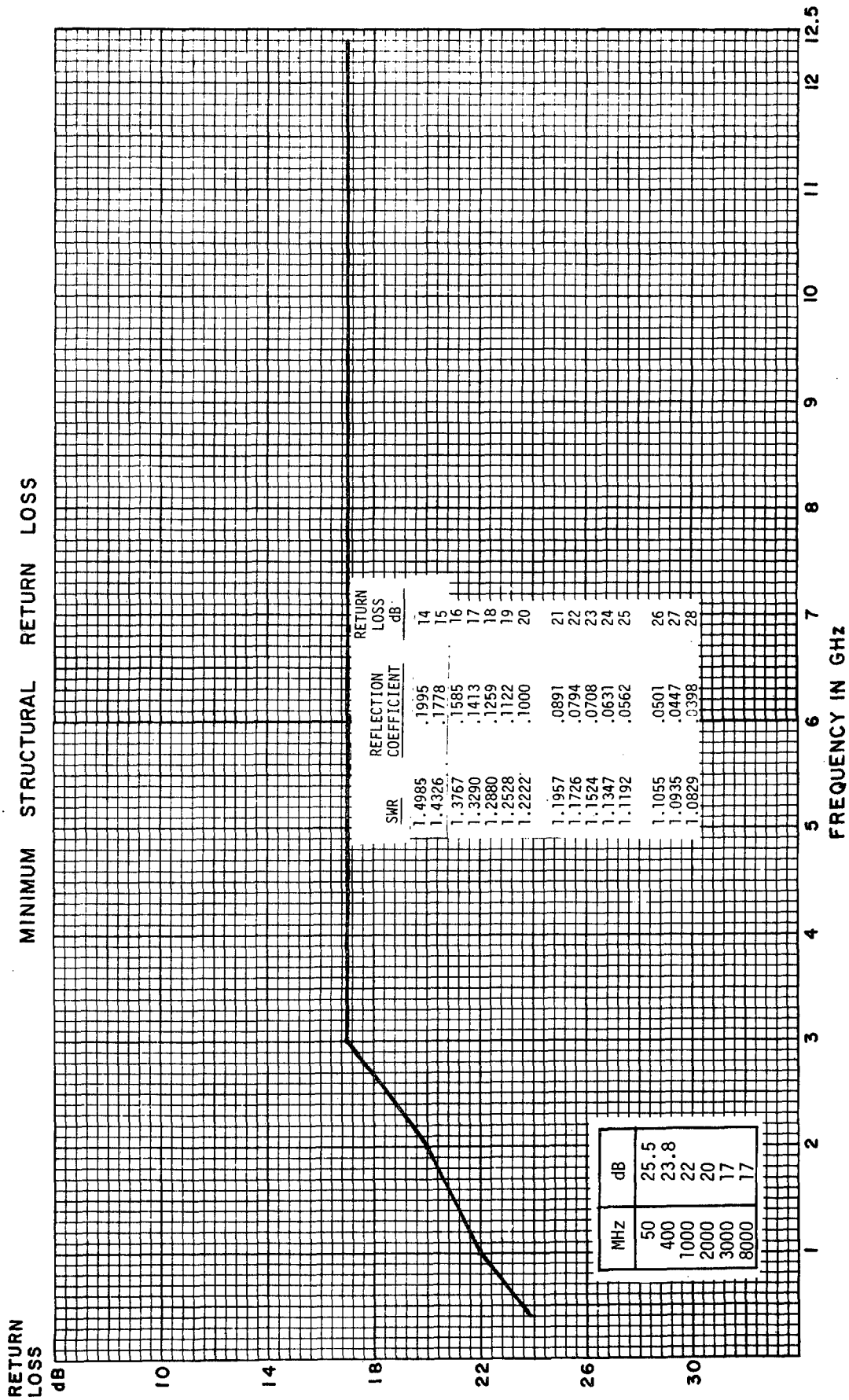
Tabulated values are for reference only. The values on the chart represent the requirements.

Maximum attenuation at 25°C, sea level ——+——+——

Maximum power at 25°C, sea level ————————

FIGURE 2. Power rating and attenuation.

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FIGURE 3. Structural return loss.

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Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Contamination: Not applicable.

Part number: M17/128-RG400.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH, EC
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - ME, AT
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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